

PRATEEK SINGH

Kanpur, Uttar Pradesh, 208027

☎ +91-8081959738 ✉ prateeksing01@gmail.com [linkedin.com/in/prateek-singh-](https://www.linkedin.com/in/prateek-singh-) [Leetcode Profile](#)

Education

Thapar Institute of Engineering and Technology, Patiala <i>Bachelor of Engineering (B.E.) in Electronics and Computer Engineering</i>	2022 – Present <i>CGPA: 8.17/10</i>
KDMA International School, Kanpur <i>XII/Central Board of Secondary Education</i>	2020 – 2021 <i>Percentage: 96%</i>
KDMA International School, Kanpur <i>X/Central Board of Secondary Education</i>	2018 – 2019 <i>Percentage: 96%</i>

Technical Skills

Technologies: Web Development, AI/ML, Cloud (AWS)

Languages: C/C++, Python, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks: React.js, Django, FastAPI

Dev Tools: Git/GitHub, VS Code, Google Collab, Anaconda, Postman, Excel, MATLAB

Relevant Courses: C, Object Oriented Programming, DBMS, Data Structures Algorithms, Data Science, Computer and Communication Networks, Operating System, Artificial Intelligence

Work Experience

Smart Mess Management System <i>Student Researcher (Experiential Learning Centre, TIET)</i>	May–Jul '25
- Built meal-wise attendance forecasting using contextual and temporal features on 10000+ records. - Computed dynamic menu score from aggregated feedback to model food-preference impact on attendance. - Benchmarked regression models; selected Random Forest for nonlinear attendance prediction. - Developed rule-based NLP pipeline to extract food-quality descriptors from student feedback. - Created role-based dashboards: student analytics and chef demand & feedback insights (Django backend).	

Projects

Hotel Reservation Platform <i>Django, React, PostgreSQL</i> GitHub	Apr–May '25
- Built a RESTful booking system with 30+ APIs for hotel management, inventory tracking, and reservations. - Designed 7+ relational database models with indexed foreign keys for integrity and efficient queries. - Implemented transaction-safe booking logic to prevent double booking under concurrent requests. - Developed JWT-based authentication with role-based access control for Admin and User workflows. - Engineered a state-driven booking lifecycle with integrated payment handling and validation checks.	

Advanced Time Table Management System for College <i>Django, React, WebSockets, MySQL</i> GitHub	Jan–Feb '25
- Designed and implemented a full-stack scheduling system to manage class timetables for students and faculty at scale. - Built a rule-based conflict detection and resolution logic to handle room, faculty, and time-slot clashes efficiently. - Integrated real-time updates using WebSockets to reflect schedule changes instantly across users. - Optimized scheduling workflows to reduce manual intervention and improve coordination efficiency.	

OS Context Switching Simulator <i>C Programming</i> GitHub	Nov–Dec '24
- Developed a user-space OS context switching simulation modeling process states and CPU scheduling. - Implemented custom stack and queue structures to manage Process Control Blocks and register state. - Simulated preemptive scheduling and validated process transitions through structured test scenarios. - Tracked execution metrics such as waiting time and turnaround time to analyze scheduling behavior.	

Extra-Curriculars

- Scholarship:** Awarded Merit Scholarship at TIET for being in the top 5% of first-year students academically.
- Publications:** Research paper (filed) and patent application (filed) on Smart Mess Management (AI + IoT) system; review process currently ongoing.
- Sports:** Secured 3rd Position in the District Chess Championship (Individuals), U-15 in Kanpur, Uttar Pradesh